# LOGI FREIGHT: PRE-TRAINING DATA & FUNCTION SPECIFICATION

Comprehensive tabular record of all raw inputs, figures found, preprocessing modules, mathematical functions, snapshot logic, and feature engineering performed prior to model training.

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## 1. Raw Dataset Figures & Parquet Ingestion Inventory

| Metric / Property | Exact Figure Found in Dataset | Technical Logic & Role |
|---|---|---|
| **Total Event Stream Rows** | **102,755,556 rows** | Total raw GPS & manifest telemetry rows across 19 continuous Parquet files. |
| **Total Parquet Files** | **19 files** (4.68 GB compressed) | Continuous chronologically ordered event stream partitioned by date. |
| **Total Unique Consignments** | **23,944 shipments** | Multi-day long-haul FTL/LTL freight journeys across India. |
| **Historical Date Range** | **July 2023 – September 2025** (27 months) | Covers all seasonal monsoons, regional traffic, and festive shipping demand cycles. |
| **Events per Consignment** | Mean: **15.6**, Median: **12.5**, Max: **34** | Each consignment contains a multi-milestone trajectory spanning origin to destination. |
| **Train Split Figures** | **32,988 snapshots** (8,361 consignments) | Files 01–10 (July 2023 – Dec 2024). Base model fitting partition. |
| **Validation Split Figures** | **14,989 snapshots** (3,781 consignments) | Files 11–14 (Jan 2025 – Apr 2025). Used for probability calibration & threshold tuning. |
| **Holdout Test Split Figures** | **15,027 snapshots** (3,802 consignments) | Files 15–19 (May 2025 – Sep 2025). Strictly untouched benchmark evaluation. |
| **Cross-Split Overlap** | **0 consignments (0.0%)** | Strict zero-leakage temporal split; no consignment appears in more than one partition. |

## 2. Raw Schema Columns: Figures Found & Engineering Logic

| Raw Input Column | Type | Null % | Engineering Action Taken | Why Was It Chosen or Dropped? (Exact Logic) |
|---|---|---|---|---|
| **Consignment No** | int64 | 0.0% | Group key; **Dropped from X** | Used to group events into life streams; dropped from X to prevent shipment ID memorization. |
| **dispatch_date** | string | 0.0% | Parsed to UTC datetime | **Anchor for Elapsed_Hours.** Physical vehicle departure from origin warehouse. |
| **delivery_date** | string | 14.1% | **Strictly Masked from X** | **CRITICAL LEAKAGE GUARD.** Used ONLY to compute Target_Is_Delayed post-prediction. |
| **original_eta** | string | 0.4% | Parsed to UTC datetime | **Anchor for Planned_Hours & Required_Speed.** The promised customer SLA deadline. |
| **lat / lng** | float64 | 58.0% | Clipped & Imputed | Vehicle waypoint coords. Imputed via corridor centroids when missing; anchor for distance & APIs. |
| **consignor_city/state** | string | 0.0% | Cleaned + Ordinal Encoded | Origin hub location. Maps geographic corridor baseline transit expectations. |
| **consignee_city/state** | string | 0.0% | Cleaned + Ordinal Encoded | Destination hub location. Defines target arrival point and distance calculations. |
| **BA_Name** | string | 0.4% | Encoded + Historical Prior | Transporter carrier name. Used to compute BA_Historical_Delay_Rate (carrier reliability). |
| **Customer Name / Shipper** | string | 0.0% | Aliased to Shipper | Client booking freight. Used to compute Shipper_Historical_Delay_Rate (dock loading speed). |
| **Vehicle Number** | string | 0.4% | **Dropped from X** | License plate. Dropped to prevent vehicle ID overfitting; carrier specification preserved via BA_Name. |
| **Vehicle Type** | string | 79.7% | Ordinal Encoded (-1 unknown) | Container spec (e.g., 20 FT Container vs 32 FT Multi-Axle). Dictates road speed ceilings. |
| **event_type** | string | 0.0% | Mapped to Stage | Operational status (IN_TRANSIT, DISPATCHED, MANIFEST). Informs journey progression phase. |
| **event_created_at** | string | 0.0% | Parsed to UTC datetime | **Anchor for As_Of Cutoff.** Defines the exact point in time when the prediction occurs. |
| **description** | string | 10.7% | **Dropped from tabular X** | Unstructured operator text. Replaced by quantitative telemetry freshness & speed indicators. |

## 3. Pre-Training Python Modules (.py Files) Created & Their Purpose

| Module Path (.py) | Module Role | Inputs Accepted | Outputs Produced | Exact Pre-Training Purpose & Logic |
|---|---|---|---|---|
| **src/data_validation.py** | Schema Validator | Raw DataFrame | bool, missing_cols | Enforces strict 26-column input contract; aliases Customer Name to Shipper and checks data types. |
| **src/data_preprocessing.py** | Stream Sanitizer | Raw DataFrame | Clean DataFrame | Parses timestamps to ISO datetimes, clips coordinates to India bounds (8-37N, 68-97E), sorts chronologically. |
| **src/leakage_prevention.py** | Leakage Guard | Event DF, As_Of | None (Raises Error) | Asserts max(t_event) <= As_Of and asserts delivery_date is absent from X. Fails loudly on any leakage. |
| **src/snapshot_generator.py** | Snapshot Reconstructor | Consignment DF | 4 Snapshot rows | Reconstructs journey timeline and extracts 4 point-in-time in-transit snapshots at 25%, 50%, 75%, 90% of route. |
| **src/target_definition.py** | Target Generator | disp, deliv, eta, as_of | Dict of 8 targets (Y) | Computes Target_Is_Delayed (0/1), Target_Remaining_Hours, Target_Delay_Hours, and P10/P50/P90 targets. |
| **src/feature_engineering.py** | Feature Engine | Event, As_Of, disp, eta | 41-feature dict (X) | Computes Elapsed_Frac, Required_Speed, Effective_Speed, Behind_Pace, Remaining_KM, and Telemetry Freshness. |
| **src/external_features.py** | Signal Fusion | lat, lng, tomtom_key | 11 external features | Fuses Open-Meteo weather (rain/temp/fog), TomTom flow speed ratios, and GDELT civic alerts into feature vector. |
| **src/weather_openmeteo.py** | Weather Connector | lat, lng | Weather dict | Queries live Open-Meteo API (or historical reanalysis) for rain mm, temp C, wind kmph, severe & fog flags. |
| **src/traffic_tomtom.py** | Traffic Flow Engine | lat, lng, api_key | Traffic dict | Calculates speed ratio (v_curr / v_freeflow) and severity tier (LOW/MODERATE/SEVERE) from empirical/live flow. |
| **src/civic_gdelt.py** | Disruption Monitor | lat, lng | Civic alert dict | Geofenced regional incident monitor checking active highway blockades within a 50 km corridor buffer. |
| **src/historical_features.py** | Priors Engine | BA, Shipper, Origin, Dest | 4 prior features | Computes expanding-window historical delay rates and volumes for carriers (BA_Name), shippers, and OD lanes. |
| **src/utils.py** | Spatial & Config | Coordinates, YAML | Distances, Config dict | Implements Haversine distance formula d = 2R arcsin(...) and YAML configuration loader. |

## 4. Pre-Training Functions Catalog & Exact Mathematical Logic

| Function Name | Source File | Mathematical Formula / Logic | Why Was This Logic Chosen? |
|---|---|---|---|
| **haversine_distance()** | src/utils.py | `d = 2R * arcsin(sqrt(sin²(Δϕ/2) + cos ϕ1 cos ϕ2 sin²(Δλ/2)))` where `R = 6371.0 km` | Provides true spherical great-circle distance between GPS coordinates without Euclidean distortion. |
| **clean_event_stream()** | src/data_preprocessing.py | `lat = clip(lat, 8.0, 37.0); lng = clip(lng, 68.0, 97.0)` `sort_values(['Consignment No', 'event_created_at'])` | Eliminates corrupted hardware GPS coordinates and ensures monotonic chronological ordering. |
| **validate_input_schema()** | src/data_validation.py | `if 'Customer Name' in df: df['Shipper'] = df['Customer Name']` `missing = [c for c in REQ_COLS if c not in df.columns]` | Ensures 100% schema compliance across varying enterprise booking formats before ingestion. |
| **verify_point_in_time_safety()** | src/leakage_prevention.py | `if (past_events['event_created_at'] > As_Of).any():` `raise LeakageViolationError(...)` | Hard runtime assertion guaranteeing zero future event lookahead in the training matrix. |
| **calculate_ground_truth_targets()** | src/target_definition.py | `Target_Is_Delayed = 1 if (t_deliv > t_eta) else 0` `Target_Remaining_Hours = (t_deliv - As_Of) / 3600` `Target_Delay_Hours = max(0, (t_deliv - t_eta) / 3600)` | Strict mathematical definition of ground truth outcomes computed purely from post-As_Of events. |
| **extract_point_in_time_features()** | src/feature_engineering.py | `Elapsed_Frac = (As_Of - t_disp) / (t_eta - t_disp)` `Req_Speed = Remaining_KM / max(0.1, Hours_Left)` `Behind_Pace = 1 if (Elapsed_Frac > 1.0 or Req_Speed > 60) else 0` | Captures physical travel dynamics, schedule deficits, and speed stress relative to contractual SLA. |
| **get_fused_external_context()** | src/external_features.py | `Traffic_Ratio = clip(v_curr / max(1, v_freeflow), 0.2, 1.0)` `Severe_Flag = 1 if (Rain > 10mm or Wind > 40km/h) else 0` | Quantifies environmental and road resistance factors that degrade vehicle transit speed. |
| **generate_consignment_snapshots()** | src/snapshot_generator.py | `for r in [0.25, 0.50, 0.75, 0.90]:` `As_Of = t_disp + (t_deliv - t_disp) * r` | Samples 4 stratified lifecycle milestones per trip (Early, Mid, Late, Approaching) for balanced training. |

## 5. Complete 41-Feature Specification Built Prior to Training

| Feature Name | Type | Source Input Field / API | Mathematical Calculation / Transformation |
|---|---|---|---|
| **Stage** | Categorical | event_type | Ordinal Encoded operational stage (IN_TRANSIT, DISPATCHED, MANIFEST). |
| **Shipper** | Categorical | Customer Name / Shipper | Ordinal Encoded client booking freight (Marico, HUL, Skoda, etc.). |
| **BA_Name** | Categorical | BA_Name | Ordinal Encoded transporter carrier company. |
| **Vehicle_Type** | Categorical | Vehicle Type | Ordinal Encoded body spec (Container 6.5 MT 20 FT, Multi-Axle, etc.). |
| **Origin_City / State** | Categorical | consignor_city/state | Ordinal Encoded freight origin hub location. |
| **Dest_City / State** | Categorical | consignee_city/state | Ordinal Encoded freight destination hub location. |
| **Event_Source / Eventable** | Categorical | Event_Source / eventable_type | Ordinal Encoded telemetry reporting source (GPS, SIM, Manifest). |
| **Traffic_Severity** | Categorical | Traffic_Congestion_Ratio | Binned tier: LOW (ratio >= 0.80), MODERATE (0.60-0.80), SEVERE (< 0.60). |
| **Civic_Disruption_Severity** | Categorical | GDELT Regional Buffer | Binned tier: NONE (0 alerts), MODERATE, HIGH. |
| **Journey_Stage_Group** | Categorical | Elapsed_Frac | Stratified bin: Early_Journey (15-35%), Mid_Transit (40-60%), Late_Journey (65-80%). |
| **Latitude / Longitude** | Numerical | lat / lng | Validated float coordinates bounded to India box. |
| **Elapsed_Hours** | Numerical | dispatch_date, As_Of | `(As_Of - dispatch_date) / 3600 (Hours elapsed since departure).` |
| **Planned_Hours** | Numerical | dispatch_date, original_eta | `(original_eta - dispatch_date) / 3600 (Total SLA duration allocated).` |
| **Elapsed_Frac** | Numerical | Elapsed_Hours, Planned_Hours | `Elapsed_Hours / Planned_Hours (Proportion of SLA time consumed).` |
| **Remaining_KM** | Numerical | lat, lng, Dest Centroid | Haversine great-circle distance from current location to destination hub. |
| **Traversed_KM** | Numerical | lat, lng, Origin Centroid | Haversine great-circle distance from origin hub to current location. |
| **Route_Progress_Ratio** | Numerical | Traversed_KM, Remaining_KM | `Traversed_KM / (Traversed_KM + Remaining_KM).` |
| **Required_Speed_Kmph** | Numerical | Remaining_KM, original_eta | `Remaining_KM / max(0.1, (original_eta - As_Of)/3600).` |
| **Effective_Speed_Kmph** | Numerical | Traversed_KM, Elapsed_Hours | `Traversed_KM / Elapsed_Hours (Historical average cruising speed).` |
| **Behind_Pace** | Numerical (Binary) | Elapsed_Frac, Req_Speed | `1 if (Elapsed_Frac > 1.0 or Required_Speed_Kmph > 60.0) else 0.` |
| **Pings_Last_3H / 6H** | Numerical | event_created_at stream | Count of active GPS pings received in the preceding 3 and 6 hours. |
| **Telemetry_Freshness_Min** | Numerical | event_created_at, As_Of | Minutes elapsed since the most recent GPS ping (detects blackouts). |
| **Hour_Of_Day / Day / Month** | Numerical | As_Of timestamp | Temporal integer components: Hour (0-23), Day (0-6), Month (1-12). |
| **BA_Historical_Delay_Rate** | Numerical | BA_Name, historical logs | Expanding historical delay probability of this carrier (default 0.28). |
| **Shipper_Historical_Delay_Rate** | Numerical | Shipper, historical logs | Expanding historical delay probability of this shipper (default 0.25). |
| **Corridor_Baseline_Hours** | Numerical | Origin-Dest historical logs | Historical empirical median duration on this OD corridor (default 24.0h). |
| **Weather_Rain_MM** | Numerical | lat, lng / Open-Meteo | Current precipitation rate in millimeters. |
| **Weather_Temp_C / Wind_Kmph** | Numerical | lat, lng / Open-Meteo | Ambient temperature in Celsius and sustained wind speed in km/h. |
| **Weather_Severe / Fog_Flag** | Numerical (Binary) | Open-Meteo weather codes | Binary indicators: Severe (Rain > 10mm or Wind > 40km/h); Fog (Code 45/48). |
| **Traffic_Congestion_Ratio** | Numerical | lat, lng / TomTom Flow | `clip(v_current / v_freeflow, 0.2, 1.0) (Highway speed ratio).` |
| **Civic_Disruption_Alert** | Numerical (Binary) | lat, lng / GDELT | Binary indicator (1 if active strike/blockade within 50 km corridor buffer). |

## 6. External APIs: Historical Training vs. Live Serving Engineering

| External Provider | Historical Pre-Training Strategy (2023-2024) | Live Production Serving Strategy (predict_live.py) | Fallback Mechanism |
|---|---|---|---|
| **Open-Meteo Weather** | Hourly historical meteorological reanalysis aligned to historical waypoint (lat, lng) and As_Of timestamp. | Direct HTTP REST API query (api.open-meteo.com/v1/forecast) returning real-time precipitation, temp, and wind. | Reverts to regional baseline (Rain=0.0mm, Temp=28C, Wind=12km/h) on network timeout. |
| **TomTom Traffic Flow** | **Empirical Speed Ratios:** Computed from historical LF GPS telemetry as v_empirical / v_freeflow (no synthetic API claims). | Live TomTom Traffic Flow API connector (flowSegmentData endpoint) querying currentSpeed and freeFlowSpeed. | Reverts to calibrated corridor baseline index (0.88 speed ratio, LOW severity) when key omitted. |
| **GDELT Civic Alerts** | Geofenced regional civil unrest and highway agitation events matched within a 50 km corridor radius buffer. | Live regional incident monitor checking active corridor blockades (emits 0 false alarms when roads are peaceful). | Defaults to Active_Alert=0, Severity=NONE. |